

TABLE NAME	CONTENT	ATTRIBUTE NAME	DATA TYPE	FORMAT	RANGE	CONSTRAINT	REFERENCE	DEFAULT	REQUIRED
USERS	Unique system ID	user_id	INT	Digits		Primary key Auto increment			Yes
	Chosen login name	user_name	VARCHAR(100)	XXXXXXXX		Not null, Unique			Yes
	Contact	email	VARCHAR(100)	XXXXXXXX		Not null, Unique			Yes
	Encrypted password	password_hash	VARCHAR(255)	XXXXXXXX		Not null			Yes
	Random hash password	salt	VARCHAR(64)	XXXXXXXX		Not null			Yes
	Given name	first_name	VARCHAR(100)	XXXXXXXX		Not null			Yes
	Family name	last_name	VARCHAR(100)	XXXXXXXX		Not null			Yes
	Contact number	phone_number	VARCHAR(20)	XXXXXXXX		Not null			Yes
	Avatar image path	profile_picture_url	VARCHAR(2048)	XXXXXXXX		Not null			Yes
	Account state	status	VARCHAR(20)	XXXXXXXX		Not null			Yes
	Source IP address	last_login_ip	VARCHAR(45)	XXXXXXXX		Null			No
	Last login timestamp	last_login_at	DATETIME	YYYY-MM-DD HH:MM:SS	1000-01-01 00:00:00 - 9999-12-31 23:59:59	Null			No
	Registration timestamp	created_at	DATETIME	YYYY-MM-DD HH:MM:SS	1000-01-01 00:00:00 - 9999-12-31 23:59:59	Not null			Yes
ROLES	Last edit timestamp	updated_at	DATETIME	YYYY-MM-DD HH:MM:SS	1000-01-01 00:00:00 - 9999-12-31 23:59:59	Not null			Yes
	Soft delete timestamp	deleted_at	DATETIME	YYYY-MM-DD HH:MM:SS	1000-01-01 00:00:00 - 9999-12-31 23:59:59	Null			No
	Unique role ID	role_id	INT	Digits		Primary key Auto increment			Yes
	Role short name	role_name	VARCHAR(50)	XXXXXXXX		Not null			Yes
	Role explanation	description	VARCHAR(200)	XXXXXXXX		Null			No
PERMISSIONS	Creation timestamp	created_at	DATETIME	YYYY-MM-DD HH:MM:SS	1000-01-01 00:00:00 - 9999-12-31 23:59:59	Not null			Yes
	Last edit timestamp	updated_at	DATETIME	YYYY-MM-DD HH:MM:SS	1000-01-01 00:00:00 - 9999-12-31 23:59:59	Not null			Yes
	Unique permission ID	permission_id	INT	Digits		Primary key Auto increment			Yes
	Unique action name	permission_name	VARCHAR(100)	XXXXXXXX		Not null			Yes
	Target module object	resource	VARCHAR(100)	XXXXXXXX		Not null			Yes
	Allowed operation	action	VARCHAR(50)	XXXXXXXX		Not null			Yes
USER_ROLES	What it grants	description	VARCHAR(200)	XXXXXXXX		Null			No
	Addition timestamp	created_at	DATETIME	YYYY-MM-DD HH:MM:SS	1000-01-01 00:00:00 - 9999-12-31 23:59:59	Not null			Yes
	Which user	user_id	INT	Digits		Primary key (user_id, role_id) Foreign Key#1	USERS(user_id)		Yes
	Which role	role_id	INT	Digits		Foreign Key#2 On delete restrict On update cascade	ROLES(role_id)		Yes
	Assignment timestamp	assigned_at	DATETIME	YYYY-MM-DD HH:MM:SS	1000-01-01 00:00:00 - 9999-12-31 23:59:59	Not null			Yes
ROLE_PERMISSIONS	Who assigned it	assigned_by	INT	Digits		Foreign Key#3 On delete restrict On update cascade	USERS(user_id)		Yes
	Which role	role_id	INT	Digits		Primary key (role_id, permission_id) Foreign Key#1	ROLES(role_id)		Yes
	Which permission	permission_id	INT	Digits		Foreign Key#2 On delete restrict On update cascade	PERMISSIONS(permission_id)		Yes
SESSIONS	Assignment timestamp	assigned_at	DATETIME	YYYY-MM-DD HH:MM:SS	1000-01-01 00:00:00 - 9999-12-31 23:59:59	Not null			Yes
	Unique session record	session_id	INT	Digits		Primary key Auto increment			Yes
	Logged-in user	user_id	INT	Digits		Foreign Key On delete cascade On update cascade	USERS(user_id)		Yes
	Client auth token	session_token	VARCHAR(255)	XXXXXXXX		Not null			Yes
	Client IP	ip_address	VARCHAR(45)	XXXXXXXX		Not null			Yes
	Device info	user_agent	VARCHAR(255)	XXXXXXXX		Not null			Yes
	Token expiry time	expires_at	DATETIME	YYYY-MM-DD HH:MM:SS	1000-01-01 00:00:00 - 9999-12-31 23:59:59	Not null			Yes
PASSWORD_RESETS	Login timestamp	created_at	DATETIME	YYYY-MM-DD HH:MM:SS	1000-01-01 00:00:00 - 9999-12-31 23:59:59	Not null			Yes
	Unique reset request	reset_id	INT	Digits		Primary key Auto increment			Yes
	Requesting user	user_id	INT	Digits		Foreign Key On delete cascade On update cascade	USERS(user_id)		Yes
	One-time reset code	token	VARCHAR(255)	XXXXXXXX		Not null, Unique			Yes
	Token deadline	expires_at	DATETIME	YYYY-MM-DD HH:MM:SS	1000-01-01 00:00:00 - 9999-12-31 23:59:59	Not null			Yes
EMAIL_VERIFICATIONS	When applied	used_at	DATETIME	YYYY-MM-DD HH:MM:SS	1000-01-01 00:00:00 - 9999-12-31 23:59:59	Null			No
	Request timestamp	created_at	DATETIME	YYYY-MM-DD HH:MM:SS	1000-01-01 00:00:00 - 9999-12-31 23:59:59	Not null			Yes
	Unique verification	verification_id	INT	Digits		Primary key Auto increment			Yes

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	User verifying email	user_id	INT	Digits		<i>Foreign Key On delete cascade On update cascade</i>	USERS(user_id)		Yes
	One-time confirmation code	token	VARCHAR(255)	XXXXXXXX		Not null, Unique			Yes
	Confirmation timestamp	verified_at	DATETIME	YYYY-MM-DD HH:MM:SS	1000-01-01 00:00:00 - 9999-12-31 23:59:59	Null			No
	Token deadline	expires_at	DATETIME	YYYY-MM-DD HH:MM:SS	1000-01-01 00:00:00 - 9999-12-31 23:59:59	Not null			Yes
	Email sent timestamp	created_at	DATETIME	YYYY-MM-DD HH:MM:SS	1000-01-01 00:00:00 - 9999-12-31 23:59:59	Not null			Yes
AUDIT_LOGS	Unique log entry	log_id	INT	Digits		<i>Primary key Auto increment</i>			Yes
	Who performed action	user_id	INT	Digits		<i>Foreign Key On delete cascade On update cascade</i>	USERS(user_id)		No
	Operation name	action	VARCHAR(50)	XXXXXXXX		Not null			Yes
	Affected table	table_name	VARCHAR(100)	XXXXXXXX		Null			No
	Affected row ID	record_id	INT	Digits		Null			No
	Before-change snapshot	old_values	JSON	{ "key": "value", ... }		Null			No
	After-change snapshot	new_values	JSON	{ "key": "value", ... }		Null			No
	Source IP	ip_address	VARCHAR(45)	XXXXXXXX		Not null			Yes
	Client device info	user_agent	VARCHAR(255)	XXXXXXXX		Not null			Yes
	Action timestamp	created_at	DATETIME	YYYY-MM-DD HH:MM:SS	1000-01-01 00:00:00 - 9999-12-31 23:59:59	Not null			Yes
SYSTEM_SETTINGS	Unique setting entry	setting_id	INT	Digits		<i>Primary key Auto increment</i>			Yes
	Config key name	setting_key	VARCHAR(100)	XXXXXXXX		Not null, Unique			Yes
	Current config value	setting_value	TEXT			Not null			Yes
	Grouping tag	category	VARCHAR(50)	XXXXXXXX		Not null			Yes
	What it controls	description	VARCHAR(200)	XXXXXXXX		Null			No
	Last change timestamp	updated_at	DATETIME	YYYY-MM-DD HH:MM:SS	1000-01-01 00:00:00 - 9999-12-31 23:59:59	Not null			Yes